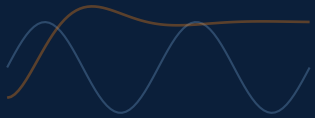
DAY 4 / 5

Mechatronics: Electrical Machines, Control, Servo Systems and Sensing



Servo Systems: Drives, FOC, Control Loops, Tuning and Compensation

3.2 PWM & limits ▪ 3.3 Field-oriented control ▪ 3.4 Cascade loops ▪ 3.5 Tuning & bandwidth ▪ 3.6 Disturbance compensation

Mechatronics Training Programme

Theoretical Course Series ▪ Day 4 of 5

Agenda and learning objectives

Code	Session	Net time
3.2	(cont.) PWM, dead time, voltage and current limits	0.5 h
3.3	Field-oriented control (FOC) of the PM motor	1.5 h
3.4	Current, velocity and position loops	1.5 h
3.5	Loop tuning and bandwidth	1.0 h
3.6	Disturbances and compensation	0.5 h
	Total (+ 10-min breaks about every hour)	**5.0 h**

One reference motor all day (permanent-magnet servo motor): $p = 4$ pole pairs, $R = 0.5\,\Omega$, $L = 3\,\text{mH}$, magnet flux $\psi_{\text{PM}} = 0.12\,\text{Vs}$, $k_t = 0.72\,\text{Nm/A}$, $J_m = 3 \times 10^{-4}\,\text{kg m}^2$ (load $3\times$ that), $V_{\text{dc}} = 325\,\text{V}$, $\hat{I}_{\text{max}} = 20\,\text{A}$, PWM at 16 kHz.

By the end of today you can ...

- explain how PWM makes an average voltage
- explain why dead time and limits exist
- sketch the $abc \rightarrow \alpha\beta \rightarrow dq$ transforms and explain FOC
- calculate PI gains for current and velocity loops
- explain bandwidth and the phase cost of delays
- describe feedforward and disturbance compensation

What we learned so far (1/2): motors and feedback

Day 1 – the motor

- Current makes torque: $T = k_t\, i$ (Nm = Nm/A·A).
- Spinning makes a voltage (back-EMF): $E = k_e\, \omega$.
- Each winding is an R–L circuit.
- Newton for rotation: $J\,\dot{\omega} = T - T_L$.

Day 2 – feedback control

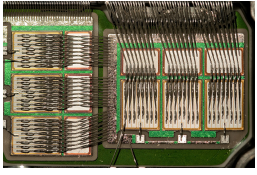
- Measure, compare, correct (cruise control).
- PID: react to the error now (P), its history (I), its trend (D).
- Bode plot: gain and phase vs. frequency; phase margin = safety distance from oscillation.



What we learned so far (2/2): the inverter, and today's chain

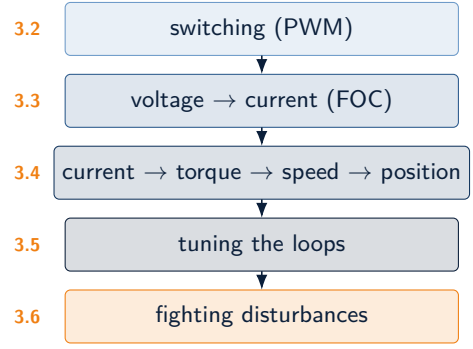
Day 3 – the inverter

- A transistor is an electronic on/off switch.
- Two switches in series = one **leg** (half-bridge): output is either $+V_{\text{dc}}$ or 0.
- Three legs feed the three motor phases from one DC voltage (**DC link**).



IGBT module with three legs' worth of switches. Pslawinski, CC BY-SA 3.0, via Wikimedia Commons

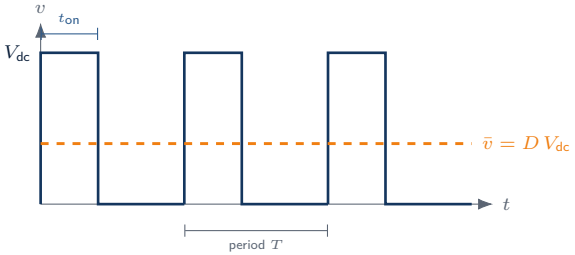
Today: from switch commands to precise motion



Session 1

3.2 (cont.) Motor Driving: PWM, Voltage and Current Limits

Refresher: duty cycle and average voltage



Analogy: flick a light switch on and off very fast – the room looks evenly *dimmed*. The motor's inductance does the same for current: like a flywheel, it smooths the pulses.

- **PWM** (pulse-width modulation): switch fully on/off at a fixed rate, vary the on-time.
- **Duty cycle** $D = t_{on}/T$ $(0 \ldots 1)$.

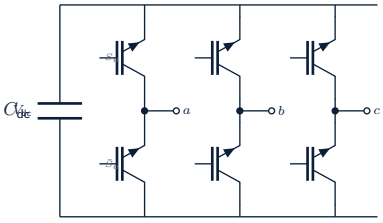
$$\bar{v} = D \cdot V_{dc}$$

Example

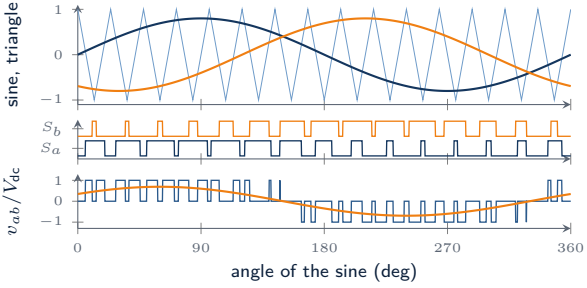
$V_{dc} = 325\,\text{V}$, $T = 62.5\,\mu\text{s}$ (16 kHz), $t_{on} = 25\,\mu\text{s}$:
$D = 25/62.5 = 0.4 \Rightarrow \bar{v} = 0.4 \cdot 325\,\text{V} = 130\,\text{V}$.

Take-away: slowly changing D gives a slowly changing average voltage – e.g. a sine.

The three-phase inverter and sinusoidal PWM

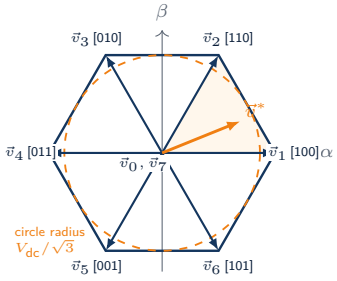


- Compare a sine with a fast triangle.
- Sine above triangle ⇒ upper switch ON.
- Duty cycle follows the sine ⇒ **average voltage is a sine**.



Peak phase voltage $\hat{V}_1 = m\,\dfrac{V_{\text{dc}}}{2}$, $m =$ **modulation index** $(0\ldots1)$. Ex.: $m = 0.8 \Rightarrow 130\,\text{V}$

Getting more voltage: space vectors and SVPWM



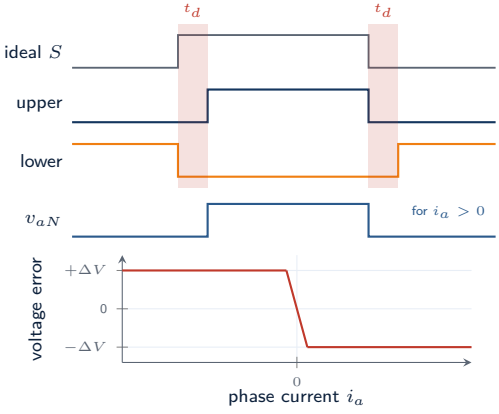
- 3 legs, each up or down: $2^3 = 8$ switch states, e.g. [100] = leg a up.
- Each state = one voltage *arrow*: 6 active + 2 zero.
- Wanted arrow $\vec{v}^*$: **mix the two neighbours in time** within one PWM period – like mixing two paints.

$\hat{V}_{1,\max}$: $V_{dc}/2$ (sine PWM) $\rightarrow$ $V_{dc}/\sqrt{3}$ (**SVPWM**) $= +15\,\%$

Example: 325 V DC link: 163 V $\rightarrow$ 188 V peak per phase.

Going further: SVPWM dwell-time algebra (volt-second balance); third-harmonic or min–max injection gives the same gain; overmodulation [Holmes & Lipo, 2003].

Dead time: why it is needed and what it costs



- Both switches of a leg ON together = short circuit of the DC link (**shoot-through**).
- So each turn-on waits a short **dead time** t_d (0.5–3 µs).
- During t_d a diode conducts: the voltage depends on the **current direction**, not on the command.

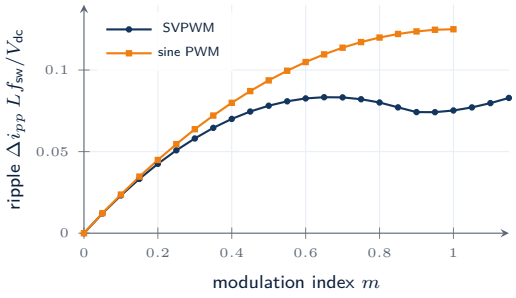
$$\Delta \bar{V} \approx t_d \, f_{\text{sw}} \, V_{\text{dc}}$$

Example

$t_d = 1\,\mu\text{s}$, $f_{\text{sw}} = 16\,\text{kHz}$, $V_{\text{dc}} = 325\,\text{V}$:
$\Delta \bar{V} = 10^{-6} \cdot 16000 \cdot 325 = 5.2\,\text{V}$ – as big as the whole back-EMF at 100 rpm!

Cure: dead-time compensation + the current loop.

Switching frequency: a trade-off



Current ripple $\Delta i_{pp} \propto \dfrac{V_{dc}}{L\, f_{sw}}$.

Example: $L = 3\,\text{mH}$, $325\,\text{V}$, $16\,\text{kHz}$, ratio 0.083:
$\Delta i_{pp} = 0.083 \cdot 325/(0.003 \cdot 16000) \approx 0.56\,\text{A}$.

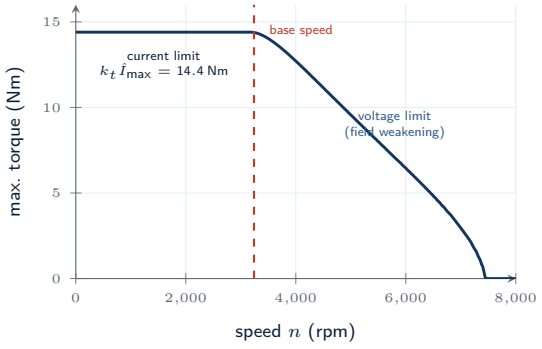
	higher f_{sw}	**lower f_{sw}**
current ripple	smaller	larger
switching loss	higher	lower
control delay	shorter	longer
noise (EMI)	more	less
audible whine	none $> 16\,\text{kHz}$	yes

Analogy: flicking the switch faster gives a smoother light, but every flick costs a little energy (heat in the transistor).

Typical servo drives: **8–16 kHz**.

[Mohan, Undeland & Robbins, 2003]

Voltage and current limits: the torque–speed envelope



Computed for the reference motor: base speed ≈ 3240 rpm.

- **Current limit**: transistors and windings get hot $\Rightarrow |i| \le \hat{I}_{\max} \Rightarrow$ maximum torque.
- **Voltage limit**: the back-EMF grows with speed; the inverter can give at most $V_{\max} = V_{\mathrm{dc}}/\sqrt{3}$.

$$E = \omega_e\,\psi_{\mathrm{PM}} \;\le\; V_{\max}$$

Example

$V_{\max} = 325/\sqrt{3} = 188\,\mathrm{V} \Rightarrow \omega_e \le 188/0.12 = 1563\,\mathrm{rad/s}$ (electrical) $\Rightarrow \le 3730$ rpm at no load; the R, L drops lower it to ≈ 3240 rpm at full torque.

Going further: voltage and current circles in the dq plane; field weakening (Session 3.3) [Morimoto et al., 1990].

Check your understanding – 3.2

Check your understanding

1. A 400 V DC link is switched with duty cycle $D = 0.25$. What is the average voltage?
2. Why can't the upper and lower switch of a leg turn on at exactly the same instant? What does the fix cost?
3. The motor stops accelerating at high speed even though the current is not at its limit. Which limit is it hitting?

Discuss with your neighbour for 2 minutes.

Summary 3.2 – PWM and limits

Key take-aways

- PWM: average voltage $= D \cdot V_{\text{dc}}$; a sinusoidal duty cycle gives a sinusoidal voltage.
- Sine PWM reaches $V_{\text{dc}}/2$, SVPWM $V_{\text{dc}}/\sqrt{3}$ ($+15\,\%$).
- Dead time prevents shoot-through but adds a voltage error.
- Higher f_{sw}: less ripple and delay, more loss.
- Current limit $\rightarrow$ max. torque; voltage limit $\rightarrow$ max. speed.

Key terms

Duty cycle – fraction of time ON.
PWM – pulse-width modulation.
Modulation index m – sine amplitude / max.
SVPWM – space-vector PWM.
Dead time – both switches OFF briefly.
Base speed – where the voltage runs out.

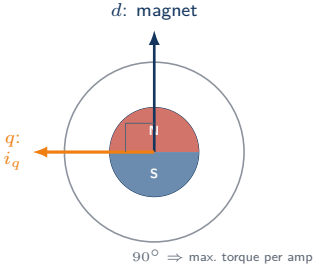
Session 2

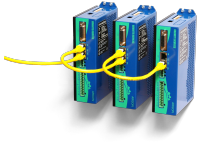
3.3 Field-Oriented Control (FOC)

Why field-oriented control? The DC-motor analogy

- Brushed DC motor: the commutator keeps the current at 90° to the magnet $\Rightarrow$ simply $T = k_t\, i$.
- PM synchronous motor (**PMSM**): the magnet is on the rotor and turns – the current must turn with it.
- Analogy: to open a door with least effort, push **perpendicular** to the door – even while it swings.
- **FOC** = electronic commutator: measure the rotor angle θ_e and split the current into
  - i_d – along the magnet (no torque),
  - i_q – 90° ahead (makes torque).

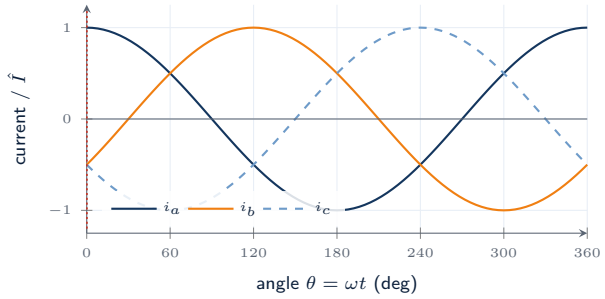
[Blaschke, 1972]; [Leonhard, 2001]; [Krishnan, 2010]





Servo drives. Tero Kontkanen, CC BY-SA 3.0, via Wikimedia Commons

Refresher: three-phase sinusoids, 120° apart



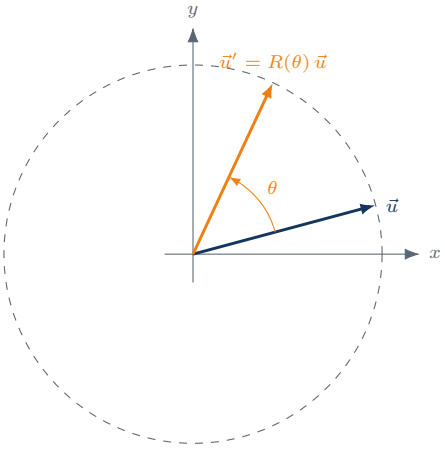
At $\theta = 0$: $i_a = 1$, $i_b = i_c = -0.5 \Rightarrow$ sum $= 0$. True at *every* instant.

$$i_a = \hat{I}\cos\theta$$
$$i_b = \hat{I}\cos(\theta - 120^\circ)$$
$$i_c = \hat{I}\cos(\theta + 120^\circ)$$

- Same amplitude, shifted by one third of a period.
- Sum is zero $\Rightarrow$ only **two** are independent.
- Coils 120° apart in space, fed 120° apart in time $\Rightarrow$ one field arrow of constant length that **rotates** at ω.

Take-away: three wiggling numbers = one rotating arrow.

Refresher: rotating vectors and rotation matrices



Length stays the same, only the direction changes.

- A vector $(x, y) =$ the complex number $x + jy$. Multiplying by $e^{j\theta}$ **rotates** it by θ.
- The same in matrix form:

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Example

Rotate $(1, 0)$ by $90°$: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Turning the *axes* by $+\theta$ instead $=$ turning the vector by $-\theta$: $(3, 4)$ seen from axes turned $90°$ reads $(4, -3)$.

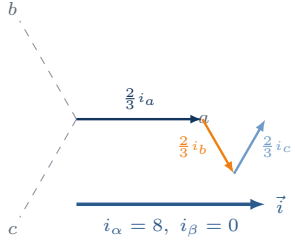
Take-away: $R(-\theta) =$ "what the vector looks like to a turned observer".

Clarke transform: three phases $\rightarrow$ two axes ($\alpha\beta$)

Clarke: $abc \rightarrow \alpha\beta$ (fixed to the stator)

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

- Each phase current pushes along its own coil axis; add the three pushes as arrows, write the result in x–y form.
- Because $i_a + i_b + i_c = 0$: $i_\alpha = i_a$, $i_\beta = (i_a + 2i_b)/\sqrt{3}$ – two current sensors suffice.
- Factor $\frac{2}{3}$: arrow length = peak phase current $\hat{I}$.

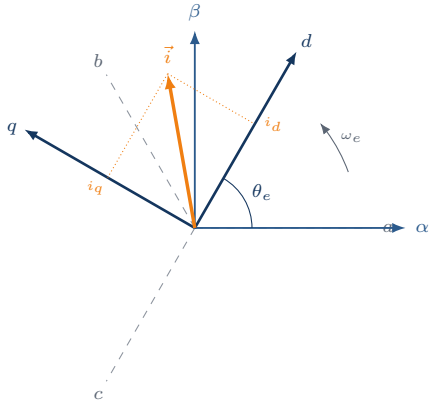


Example

$i_a = 8\,\text{A}$, $i_b = i_c = -4\,\text{A}$:
$i_\alpha = \frac{2}{3}(8 + 2 + 2) = 8\,\text{A}$,
$i_\beta = (8 - 8)/\sqrt{3} = 0\,\text{A}$.

The arrow still **rotates** at ω_e: i_α, i_β are still sines [Clarke, 1943].

Park transform: looking at the rotor from a rotating camera



d axis glued to the magnet's north pole.

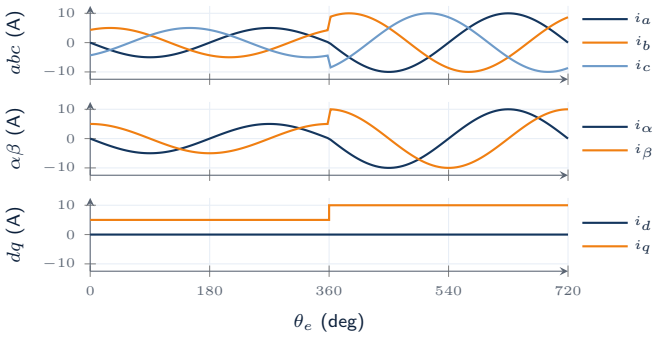
- Imagine a **camera bolted to the rotor**. In its picture the magnet stands still.
- The current arrow rotates at the same speed ⇒ it *also* looks still: two constant numbers i_d, i_q.
- Maths: rotate the axes by θ_e = rotate the vector by $-\theta_e$:

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos\theta_e & \sin\theta_e \\ -\sin\theta_e & \cos\theta_e \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}$$

Take-away: same physical arrow, new coordinates – AC becomes DC.

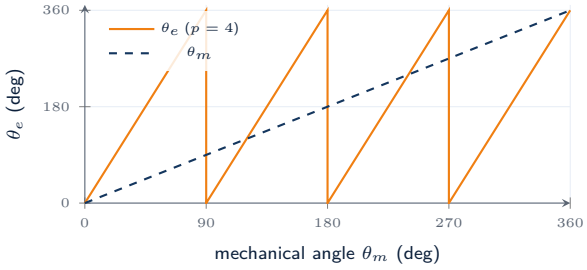
Going further: the $\frac{2}{3}$ (amplitude-invariant) scaling makes power $p = \frac{3}{2}(v_d i_d + v_q i_q)$; a power-invariant variant uses $\sqrt{2/3}$ – check the drive manual. Inverse Park uses $R(+\theta_e)$ [Park, 1929].

The same current seen in the three frames



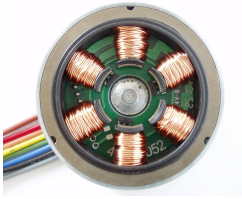
- Torque command steps from 5 A to 10 A at $\theta_e = 360°$.
- abc: three sines.
- $\alpha\beta$: two sines, same frequency.
- dq: **flat lines** – $i_d = 0$, i_q makes a clean step.
- A PI controller (Day 2) handles constants perfectly $\Rightarrow$ no error at any speed.

Electrical vs. mechanical angle: $\theta_e = p\,\theta_m$



$$\theta_e = p\,\theta_m, \qquad f_e = \frac{p\,n}{60}$$

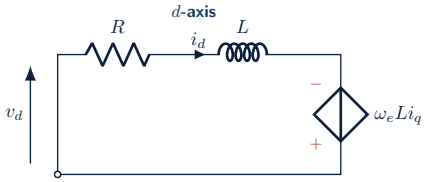
Example: $p = 4$, $n = 3000$ rpm:
$f_e = 4 \cdot 3000/60 = 200$ Hz, $\omega_e = 2\pi \cdot 200 = 1257$ rad/s.

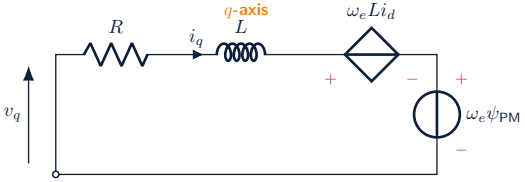


12-slot stator. Medvedev, CC BY-SA 3.0, via Wikimedia Commons

- $p =$ number of **pole pairs** (8 magnet poles $\Rightarrow p = 4$).
- One mechanical turn $= p$ electrical turns: the current must go through p full sine periods.
- An angle error of 2° mechanical $= 8^\circ$ electrical.

The motor seen by the rotating camera: two R–L circuits





$$v_d = R\,i_d + L\frac{di_d}{dt} - \omega_e L\,i_q$$

$$v_q = R\,i_q + L\frac{di_q}{dt} + \omega_e L\,i_d + \underbrace{\omega_e \psi_{\text{PM}}}_{\text{back-EMF}}$$

- Each axis = the Day-1 winding (R, L) plus speed-dependent sources; time constant $L/R = 6$ ms.
- Back-EMF appears only on q. The $\omega_e L\,i$ terms couple the axes.

Going further: derived from $\vec{v} = R\vec{i} + d\vec{\psi}/dt$ in rotating coordinates, which adds $j\omega_e\vec{\psi}$; IPM motors have $L_d \neq L_q$.

Torque comes from i_q

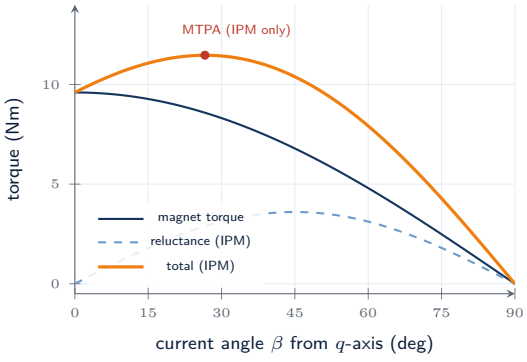
$$T_e = \tfrac{3}{2}\, p\, \psi_{\mathsf{PM}}\, i_q = k_t\, i_q$$

Example (reference motor)

$k_t = 1.5 \cdot 4 \cdot 0.12\,\mathsf{Vs} = 0.72\,\mathsf{Nm/A}$
$i_q = 10\,\mathsf{A} \Rightarrow T_e = 0.72 \cdot 10 = 7.2\,\mathsf{Nm}.$

- Set $i_d^* = 0$: all the current makes torque.
- If the current arrow is tilted by β away from q, only $\cos\beta$ of it is useful (blue curve).

Going further: in **IPM** motors ($L_q > L_d$) a negative i_d adds *reluctance torque* $\frac{3}{2}p(L_d - L_q)i_d i_q$; the best angle is **MTPA** (max. torque per amp, here $\beta = 26.6°$, +19 %). Above base speed, negative i_d weakens the magnet flux (**field weakening**) to reach higher speed [Jahns et al., 1986].



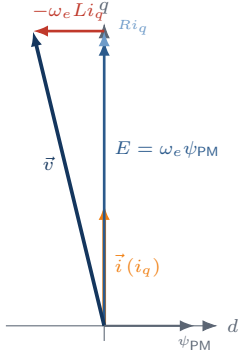
Illustrative IPM motor, $\hat{I} = 20\,\mathsf{A}$. For our surface-magnet motor only the blue curve exists: best at $\beta = 0$.

Worked example: operating point at 3000 rpm

Reference motor, $i_d = 0$, $i_q = 10\,\text{A}$, $n = 3000\,\text{rpm}$.

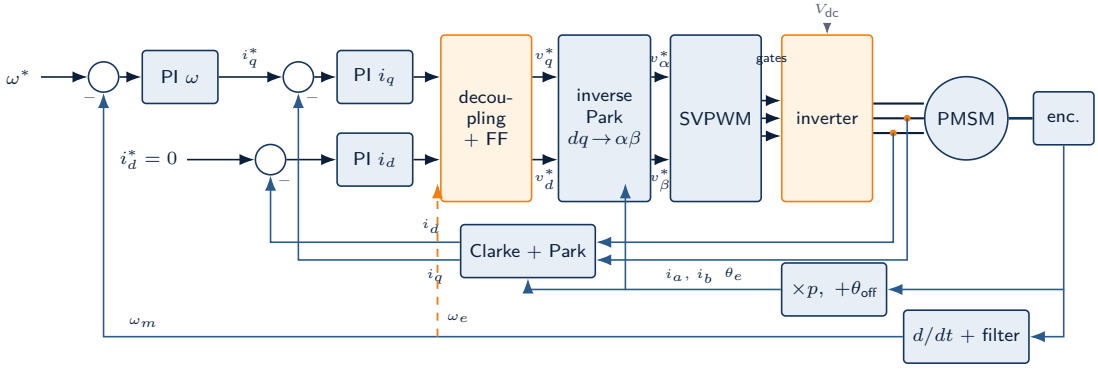
1. Electrical speed:
   $\omega_e = p \cdot \frac{2\pi n}{60} = 4 \cdot 314.2\,\text{rad/s} = 1257\,\text{rad/s}$.
2. Back-EMF: $E = \omega_e \psi_{\text{PM}} = 1257 \cdot 0.12 = 150.8\,\text{V}$.
3. Axis voltages:
   $v_q = R\,i_q + E = 5.0 + 150.8 = 155.8\,\text{V}$;
   $v_d = -\omega_e L\,i_q = -1257 \cdot 0.003 \cdot 10 = -37.7\,\text{V}$.
4. Length: $|\vec{v}| = \sqrt{37.7^2 + 155.8^2} = 160.3\,\text{V}$
   $< V_{\text{max}} = 188\,\text{V}$ ✓ (85 % used).

Torque $k_t i_q = 7.2\,\text{Nm}$; power $T\omega_m = 7.2 \cdot 314 = 2.26\,\text{kW}$.



Vector diagram to scale (40 V/cm). At higher speed or load the voltage arrow reaches the limit circle.

FOC: the basic structure



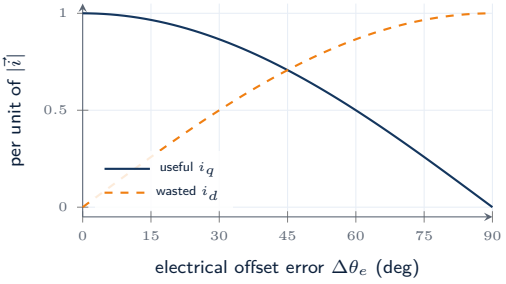
Every PWM period (62.5 µs): 1. measure i_a, i_b, angle; 2. Clarke + Park; 3. two PIs; 4. inverse Park; 5. new duty cycles.

Going further: decoupling: add the known terms $-\omega_e L i_q$ and $+\omega_e(L i_d + \psi_{\text{PM}})$ to the PI outputs, so each PI sees only its own R–L circuit.

Rotor angle alignment and offset errors

- The encoder's zero is wherever it was mounted; FOC needs $\theta_e = 0$ exactly at the magnet's d-axis.
- Wrong offset = the camera is mounted crooked: we push at the wrong angle.
- Fixes: absolute encoder with factory offset (typical); DC alignment (push a current, rotor snaps to $\theta_e = 0$); Hall sensors at start-up.

$$T_e = k_t \, |\vec{i}| \cos \Delta\theta_e$$



Example

2° mech., $p = 4 \Rightarrow 8^\circ$ el.: torque $-1\,\%$, but $14\,\%$ of the current wasted as heat. Beyond 90° the torque reverses $\Rightarrow$ **runaway**.

Check your understanding – 3.3

Check your understanding

1. $i_a = 8\,\text{A}$, $i_b = -4\,\text{A}$, $i_c = -4\,\text{A}$. What are i_α and i_β?
2. In the "rotating camera" view, why do the motor currents look constant?
3. An 8-pole motor turns at 1500 rpm. What is the electrical frequency of its currents?

Discuss with your neighbour for 2 minutes.

Summary 3.3 – Field-oriented control

Key take-aways

- Three phase currents = one rotating arrow.
- Clarke: $abc \to \alpha\beta$; Park: view from the rotor $\Rightarrow$ DC values i_d, i_q.
- Torque $T_e = k_t i_q$; keep $i_d = 0$ (surface magnets).
- Each axis is an R–L circuit; back-EMF on q.
- FOC: measure, transform, two PIs, transform back, PWM – every period. Needs a correct angle.

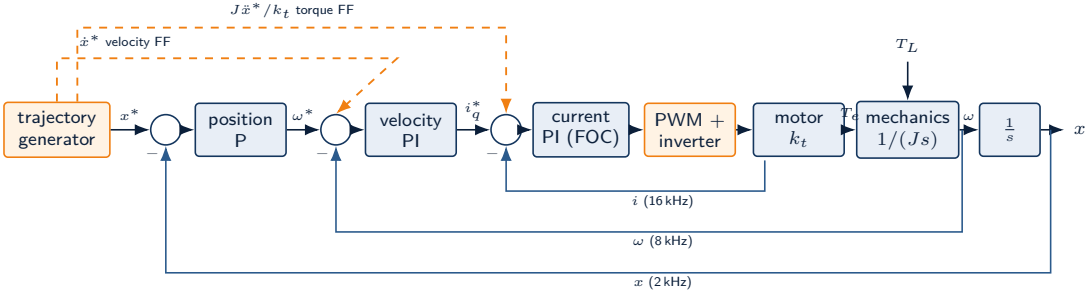
Key terms

PMSM – permanent-magnet synchronous motor.
$\alpha\beta$ **frame** – two fixed axes.
dq **frame** – two axes turning with the rotor.
i_d **/** i_q – flux / torque current.
Pole pairs p – $\theta_e = p\theta_m$.
MTPA, field weakening – see "going further".

Session 3

3.4 Current, Velocity and Position Loops

The nested (cascade) control structure



- Analogy: navigator (position) → cruise control (speed) → engine management (torque).
- Each loop treats the closed inner loop as its "actuator".
- Inner loops are **faster**; they also enforce limits (current → torque).
- Dashed: **feedforward** – known commands sent straight to the inner loops.

[Ellis, 2012]

Current-loop PI: a step-by-step recipe

- Plant (one axis): the R–L winding, $G_i(s) = \dfrac{1}{R + Ls}$, plus a small **delay** T_σ (compute + PWM).
- PI: $C_i(s) = K_p \dfrac{1 + sT_i}{sT_i}$.
- Idea: let the PI's zero cancel the slow winding pole, then pick K_p so the delay is still safe.

$$T_i = \frac{L}{R}, \qquad K_p = \frac{L}{2\,T_\sigma}$$

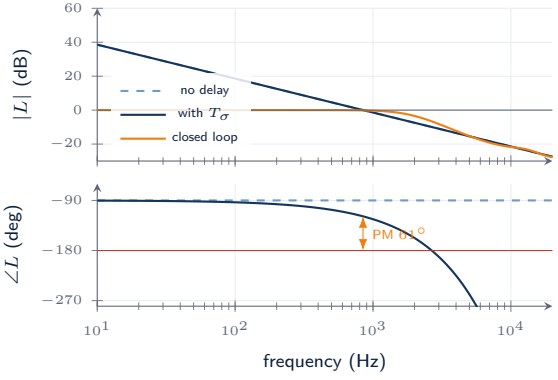
This rule is called the **technical optimum** (TO).

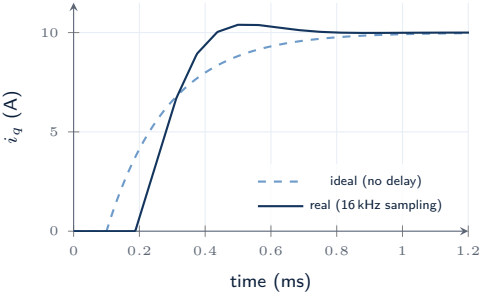
Going further: the closed loop becomes 2nd order with $\zeta = 0.707$ (4.3 % overshoot); saturation lowers L at high current, so check K_p there.

Worked example (reference motor)

1. $T_s = 1/16\,\text{kHz} = 62.5\,\mu\text{s}$; $T_\sigma = 1.5\,T_s = 93.75\,\mu\text{s}$.
2. $T_i = 3\,\text{mH}/0.5\,\Omega = 6\,\text{ms}$.
3. $K_p = \dfrac{3\,\text{mH}}{2 \cdot 93.75\,\mu\text{s}} = 16\,\text{V/A}$.
4. Crossover $\omega_c \approx K_p/L = 5333\,\text{rad/s}$ ($f_c \approx 850\,\text{Hz}$); delay costs $360^\circ \cdot 850 \cdot 93.75\,\mu\text{s} = 29^\circ \Rightarrow \text{PM} \approx 61^\circ$.

Current loop: computed frequency and step response





Step $0 \rightarrow 10$ A: current follows in ≈ 0.3 ms, overshoot only 3.9 %.
The delay bends the phase curve down at high frequency – that is what limits K_p.

Velocity-loop PI: a step-by-step recipe

- Plant: current $\rightarrow$ torque $\rightarrow$ speed. Newton: $J\dot{\omega} = k_t i_q \Rightarrow$ an **integrator** $\dfrac{k_t}{Js}$.
- Plus a small lag T_{eq}: closed current loop + speed filter.
- Choose a spacing factor a ($2 \ldots 4$): crossover a times below the lag, PI corner a times below the crossover.

$$K_{p\omega} = \frac{J\,\omega_c}{k_t}, \qquad \omega_c = \frac{1}{a\,T_{\text{eq}}}$$

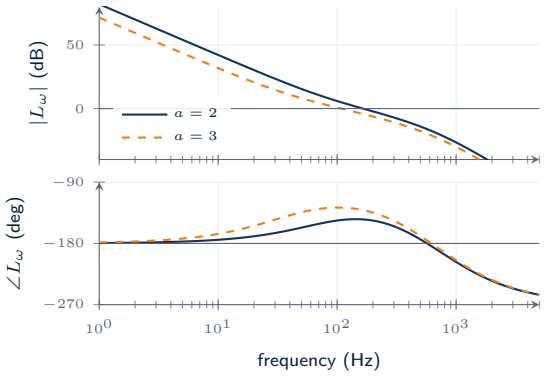
Going further: this is the **symmetric optimum** [Kessler, 1958]: $T_{i\omega} = a^2 T_{\text{eq}}$, PM $= \arctan\frac{a^2-1}{2a}$ (37° at $a=2$, 53° at $a=3$); a reference prefilter removes the large step overshoot.

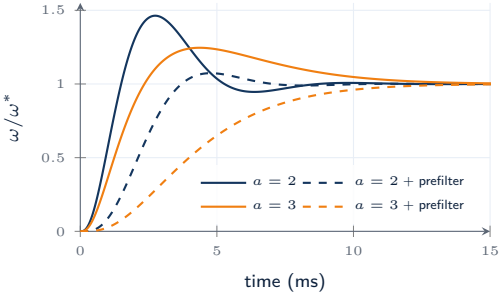
Worked example: spacing factor 2

1. $T_{\text{eq}} = 2 \cdot 93.75\,\mu\text{s} + 0.31\,\text{ms} \approx 0.5\,\text{ms}$.
2. $\omega_c = 1/(2 \cdot 0.5\,\text{ms}) = 1000\,\text{rad/s}$ (159 Hz).
3. $J = J_m(1+3) = 1.2 \cdot 10^{-3}\,\text{kg m}^2$; $K_{p\omega} = 1.2 \cdot 10^{-3} \cdot 1000/0.72 = 1.67\,\text{A s/rad}$.
4. $T_{i\omega} = a^2 T_{\text{eq}} = 2\,\text{ms}$.

Larger $a \Rightarrow$ slower but calmer ($a=3$: $f_c = 106\,\text{Hz}$).

Velocity loop: computed Bode plot and step responses





$a = 2$: 164 Hz, PM $35°$, 46 % overshoot (7 % with prefilter).
$a = 3$: 107 Hz, PM $53°$, 25 % (0 % with prefilter).
More phase margin = less overshoot.

Position loop: a P controller and the following error

- With a good velocity loop, speed $\approx$ command; position is the integral of speed.
- So a simple **P controller** reaches any fixed target with zero error. Gain k_v (1/s).
- Following a ramp (constant speed v) leaves a constant lag: the **following error**.

$$e_{\text{ss}} = \frac{v}{k_v} \qquad \text{(P loop, no feedforward)}$$

Analogy: a dog on a leash trotting behind its owner – the faster the walk, the longer the leash.

Rule: $k_v \approx$ velocity crossover / 4…10. No overshoot allowed in machining.

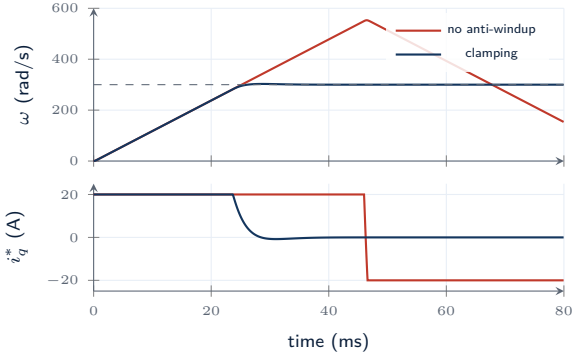
Worked example

1. Machine-tool axis: $k_v = 50\,\text{s}^{-1}$, feed $v = 10\,\text{m/min} = 167\,\text{mm/s}$.
2. $e_{\text{ss}} = 167/50 = 3.3\,\text{mm}$ (!)
3. Doubling k_v halves it – but stability limits k_v.

Two axes cutting a contour need **equal** k_v, otherwise the path is distorted – or use feedforward (two slides on).

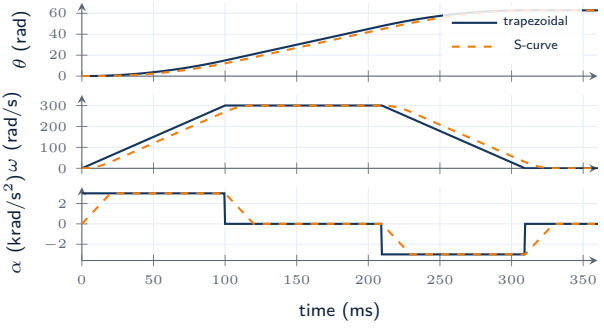
Torque and speed limits; integrator windup

- Limits: $|i_q^*| \le \hat{I}_{\max}$, $|\omega^*| \le \omega_{\max}$, plus the voltage limit.
- While the output is limited, the loop is effectively **open**; the integrator keeps adding up error.
- Result: **windup** – huge overshoot when the limit releases.
- Analogy: holding the accelerator floored and still "remembering" to push harder.
- Cure (**anti-windup**): stop integrating while saturated (clamping).



300 rad/s step, current limit 20 A: peak 553 rad/s vs. 303 rad/s.

Trajectory generation: trapezoidal vs. S-curve profiles

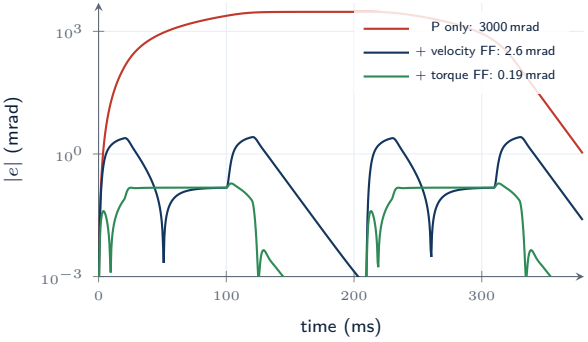


Move 10 revolutions, max. 300 rad/s, max. 3000 rad/s^2.

- **Trapezoidal**: speed ramps; acceleration *jumps* $\Rightarrow$ jolts that shake the machine. Total 309 ms.
- **S-curve**: acceleration ramps too (limited **jerk**). Total 329 ms (+6 %), much smoother.
- Analogy: a good lift driver – you never feel the start.
- Peak torque $J\alpha = 3.6\,\text{Nm}$ ($i_q = 5\,\text{A}$) – within limits.

[Biagiotti & Melchiorri, 2008]

Following error and how feedforward removes it



Simulated cascade on the S-curve move; legend: peak $|e|$. **Log scale**: each step is $\sim 1000\times$.



5-axis machining centre. G. McKechnie, CC BY-SA 2.0, via Wikimedia Commons

- Velocity FF: send the planned speed $\dot{x}^*$ directly.
- Torque FF: send the needed current $\hat{J}\ddot{x}^*/k_t$ directly.
- Feedback only fixes what is left. FF acts outside the loop: **no stability change**.

Check your understanding – 3.4

Check your understanding

1. $L = 6\,\text{mH}$, $T_\sigma = 100\,\mu\text{s}$. What current-loop gain K_p does the recipe give?
2. An axis moves at $v = 0.2\,\text{m/s}$ with $k_v = 40\,\text{s}^{-1}$ and no feedforward. Following error?
3. Why must the loops be tuned from the inside out?

Discuss with your neighbour for 2 minutes.

Summary 3.4 – Cascade loops

Key take-aways

- Cascade: position (P) $\rightarrow$ velocity (PI) $\rightarrow$ current (PI); inner = faster.
- Current PI: $T_i = L/R$, $K_p = L/(2T_\sigma) \Rightarrow \approx 850\,\text{Hz}$.
- Velocity PI: $K_p = J\omega_c/k_t$ with $\omega_c = 1/(aT_{\text{eq}})$.
- P position loop: following error v/k_v.
- Limits need anti-windup; S-curves limit jerk; feedforward removes most following error.

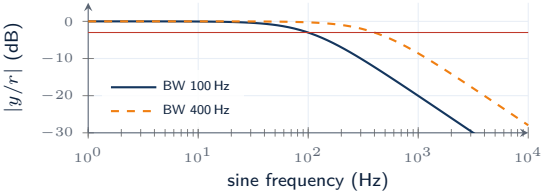
Key terms

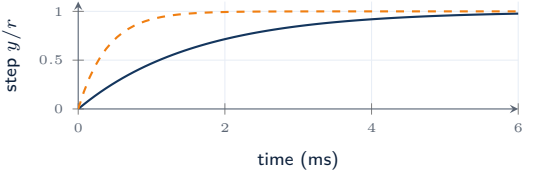
Cascade – loops nested inside each other.
Crossover ω_c – where loop gain = 1.
k_v – position-loop gain (1/s).
Following error – lag behind a moving target.
Windup – integrator overfills during a limit.
Jerk – rate of change of acceleration.
Feedforward – act on the known command.

Session 4

3.5 Loop Tuning and Bandwidth

Refresher: what does “bandwidth” mean?





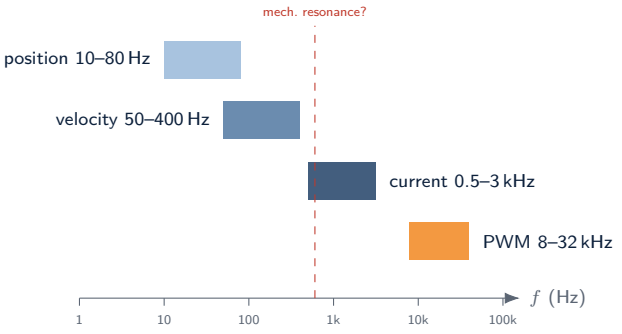
- Feed the closed loop a sine command and watch the output.
- Slow sines: followed perfectly. Fast sines: output too small and late.
- **Bandwidth** = frequency where the output drops to $70\,\%$ ($-3\,\mathrm{dB}$).
- Analogy: a dancer copying a leader – slow moves are easy, fast ones get lost.

$$\text{rise time } t_r \approx \frac{0.35}{f_{\mathrm{BW}}}$$

Example: $f_{\mathrm{BW}} = 100\,\mathrm{Hz} \Rightarrow t_r \approx 3.5\,\mathrm{ms}$; $400\,\mathrm{Hz} \Rightarrow 0.9\,\mathrm{ms}$.

Take-away: higher bandwidth = faster response.

The bandwidth hierarchy of a servo drive



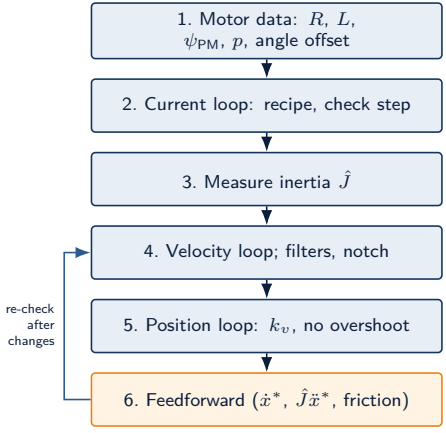
- Each loop is a few times slower than the one inside it.
- Current: limited by sampling and PWM delay. Velocity: by **mechanical resonances** and filters. Position: by the velocity loop.



Industrial robot. Oleg Yunakov, CC BY-SA 4.0, via Wikimedia Commons

Robots with elastic gears: velocity bandwidth only a few tens of Hz; stiff direct drives: several hundred Hz [Ellis, 2012].

Tuning procedure: from the inside out



- Each outer loop is designed on the **closed** inner loop – tune outer first and you tune against the wrong plant.
- Measure, don't guess: e.g. inertia from an acceleration test, $\hat{J} = k_t\,\Delta i_q / \Delta\dot{\omega}$.

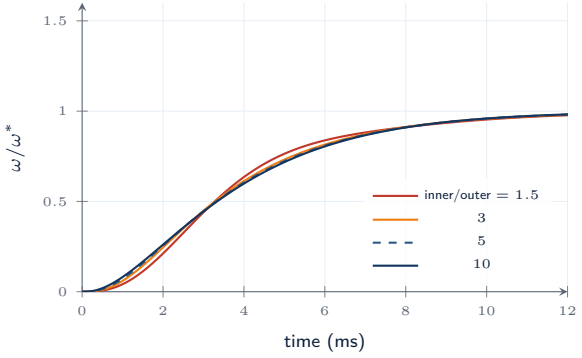
Example

$\Delta i_q = 5\,\text{A}$ gives $\Delta\dot{\omega} = 3000\,\text{rad/s}^2$:
$\hat{J} = 0.72 \cdot 5/3000 = 1.2 \times 10^{-3}\,\text{kg m}^2$.

- Raise gains until a margin limit (PM $\geq 45°$, GM $\geq 6\,\text{dB}$), then back off.
- Check large steps last (limits, anti-windup).

[Ellis, 2012]; [Åström & Hägglund, 2006]

Bandwidth separation: rule of thumb 3–10×



Computed: velocity PI designed for 1000 rad/s *assuming a perfect current loop*; the real current loop has bandwidth ω_i.

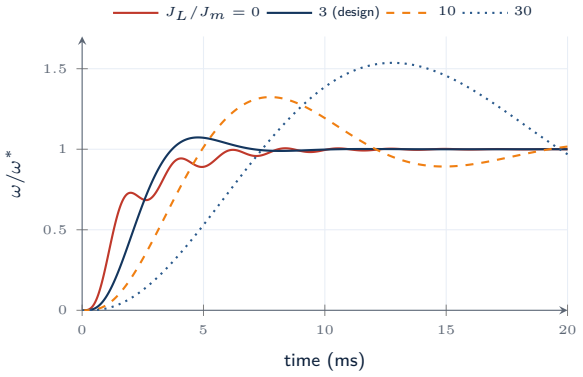
- A slow inner loop adds lag (phase loss) at the outer crossover:

$$\Delta\varphi = \arctan(\omega_c/\omega_i)$$

ω_i/ω_c	phase lost
1.5	33.7°
3	18.4°
10	5.7°

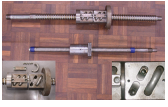
- Below 3×: ringing (red curve).
- Above 10×: safe, but the outer loop could be faster.

Effect of the load inertia ratio



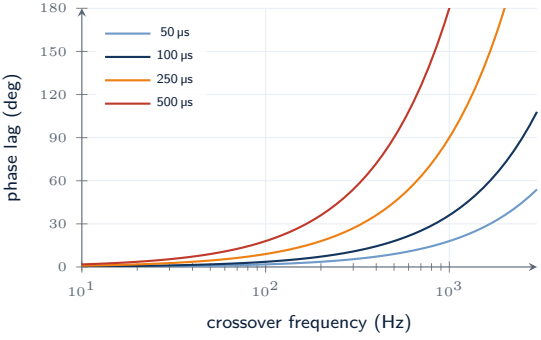
Same gains (designed for ratio 3). Crossover / PM: 0: 439 Hz/12°; 3: 164 Hz/35°; 10: 80 Hz/31°; 30: 43 Hz/21°.

- Loop gain $\propto K_{p\omega} k_t / J$: more load inertia $\Rightarrow$ **lower crossover**, sluggish; less $\Rightarrow$ higher, towards the delays, oscillating.
- Analogy: bicycle gears – the gear (or screw) ratio changes how heavy the load feels at the motor.
- Rule: retune with $\hat{J}$; aim for $J_L/J_m \leq 3 \ldots 10$.



Ball screws. Graibeard, CC BY-SA 3.0, via Wikimedia Commons

Delays and their phase cost at crossover



Analogy: a shower with a long pipe – you turn the tap, the temperature changes seconds later, so you overshoot.

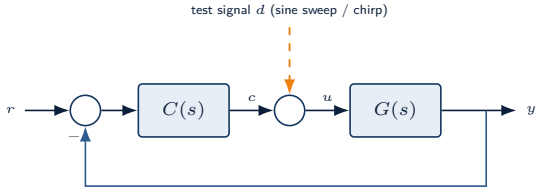
$$\Delta\varphi = 360^\circ \cdot f_c \cdot T_d$$

Delay source	typical
Computation (1 sample at 16 kHz)	62.5 µs
PWM update ($T_s/2$)	31 µs
Encoder data transfer	10–30 µs
Velocity filter	0.1–1 ms

Example

$T_d = 0.5$ ms, $f_c = 160$ Hz: $360 \cdot 160 \cdot 0.0005 = 28.8^\circ$ of phase margin lost.

Measuring frequency responses (FRFs)



plant $G(j\omega) = \dfrac{Y(j\omega)}{U(j\omega)}$ at each test frequency

- Add a small test signal *inside* the running loop – the machine stays under control.
- Compare output and input amplitude and phase at each frequency ⇒ a measured Bode plot.
- Large enough to beat friction and noise, small enough to stay linear.

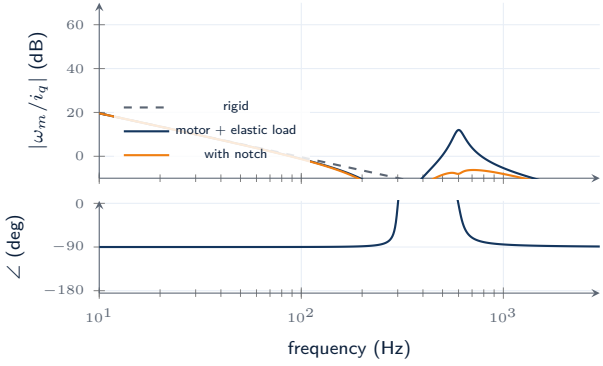


HP 3562A dynamic signal analyzer. Loxyger, CC BY 4.0, via Wikimedia Commons

Modern drives have this analyser built in: the software injects a chirp and plots the Bode diagram.

Going further: estimate $\hat{G} = \hat{S}_{uy}/\hat{S}_{uu}$ from cross-spectra and check the **coherence** $\gamma^2 \approx 1$.

A measured-style FRF: resonance and notch filter



- Motor and load joined by a springy shaft = **two-mass system**.
- Analogy: a playground swing – pushed at its own frequency, it swings huge.
- Dip (anti-resonance) at 300 Hz, peak (**resonance**) at 600 Hz.
- The peak can push the loop gain above 1 where the phase is bad ⇒ "buzz".
- **Notch filter**: cuts a narrow band around 600 Hz in the current command.

Going further: $f_a = \frac{1}{2\pi}\sqrt{k_s/J_L}$, $f_r = f_a\sqrt{1 + J_L/J_m}$; notch $N(s) = \frac{s^2+2\zeta_z\omega_r s+\omega_r^2}{s^2+2\zeta_p\omega_r s+\omega_r^2}$. More on Day 5.

Check your understanding – 3.5

Check your understanding

1. A velocity loop has a bandwidth of 200 Hz. Roughly what rise time do you expect?
2. How much phase does a 200 µs delay cost at a 250 Hz crossover?
3. The load inertia doubles but the gains stay the same. What happens to the crossover frequency?

Discuss with your neighbour for 2 minutes.

Summary 3.5 – Tuning and bandwidth

Key take-aways

- Bandwidth = how fast a loop can follow; $t_r \approx 0.35/f_{\mathrm{BW}}$.
- Tune inside out; separate bandwidths 3–10×.
- Inertia changes the loop gain: retune with $\hat{J}$.
- Every delay costs $360^\circ f_c T_d$ of phase.
- Measure FRFs in closed loop; tame resonances with notches.

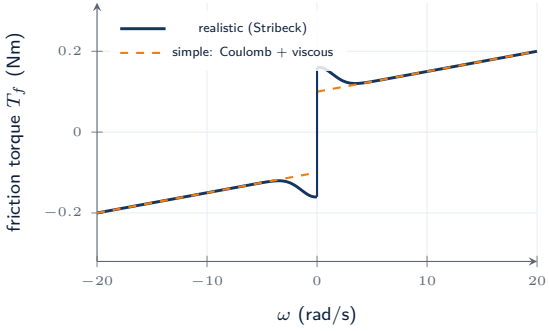
Key terms

Bandwidth – -3 dB frequency.
Phase margin – safety from oscillation.
Inertia ratio – J_L/J_m.
FRF – measured frequency response.
Resonance – springy mechanics amplifying one frequency.
Notch filter – removes a narrow band.

Session 5

3.6 Disturbances and Compensation

Disturbances: load torque and friction



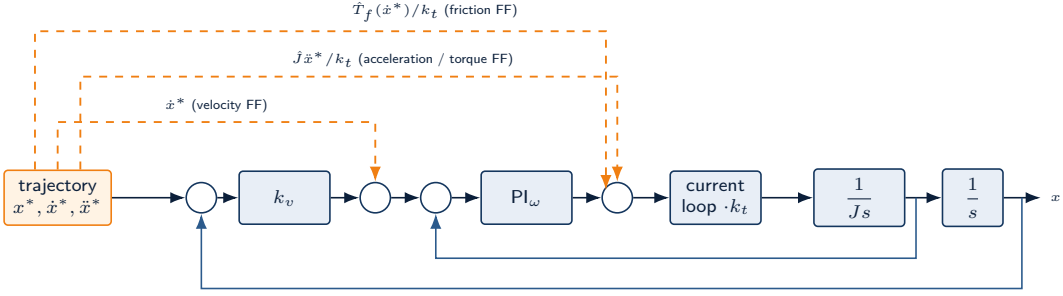
Breakaway 0.16 Nm, sliding 0.1 Nm, plus 5 mNm per rad/s.

$$J\dot{\omega} = k_t i_q - T_L - T_f(\omega)$$

- **Load torque** T_L: cutting forces, gravity on vertical axes.
- **Friction**: jumps at $\omega = 0$ (harder to start than to keep sliding) $\Rightarrow$ stick-slip, small jerks when reversing.
- The PI removes a constant T_L eventually, but a **dip** occurs first.
- Also: cogging, dead time, sensor noise.

[Armstrong-Hélouvry, Dupont & Canudas de Wit, 1994]

Feedforward: velocity, acceleration/torque and friction

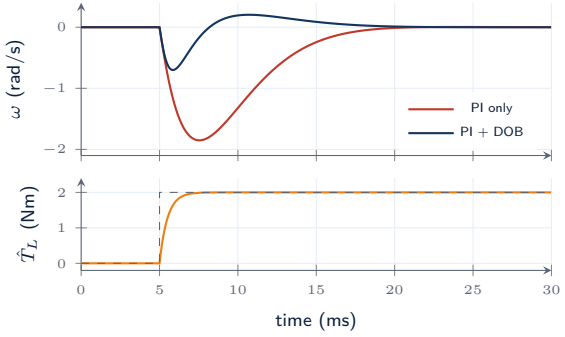


$$i_{q,\mathrm{FF}}^* = \left(\hat{J}\ddot{x}^* + \hat{T}_f\right)/k_t$$

Example: $\hat{J} = 1.2 \times 10^{-3}\,\mathrm{kg\,m^2}$, $\ddot{x}^* = 3000\,\mathrm{rad/s^2}$, $\hat{T}_f = 0.1\,\mathrm{Nm}$: $(3.6 + 0.1)/0.72 = 5.1\,\mathrm{A}$.

- Feedforward = **predict** the needed current from the plan; feedback only corrects the leftover.
- Needs smooth (S-curve) commands and a good $\hat{J}$.
- Outside the loop: no stability change.

Disturbance observer (DOB): estimate the unknown push



Simulated: holding $\omega^* = 0$, load step 2 Nm at 5 ms.

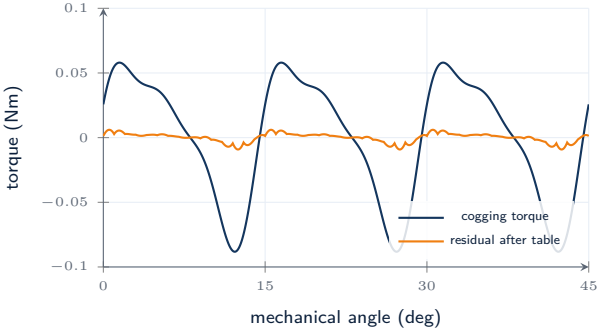
- Idea: compare what the motor *should* do ($k_t i_q$) with what it *does* ($\hat{J}\dot{\omega}$). The difference is the unknown disturbance.

$$\hat{T}_L \approx k_t\, i_q - \hat{J}\,\dot{\omega} \quad \text{(low-pass filtered)}$$

- Add $\hat{T}_L / k_t$ to the current command $\Rightarrow$ the disturbance is cancelled.
- Result: speed dip 1.85 rad/s $\rightarrow$ 0.70 rad/s (2.6$\times$ smaller).
- Bonus: $\hat{T}_L$ can detect collisions.

Going further: the filter $Q(s) = 1/(1 + s\tau_Q)$ makes the derivative usable; $1/\tau_Q \approx 0.5$–$2\times\omega_c$, limited by noise and delay [Ohnishi, Shibata & Murakami, 1996].

Cogging and torque ripple compensation



Illustrative 12-slot/8-pole motor: 0.146 Nm → 0.015 Nm peak-to-peak.

- **Cogging**: magnets "click" into the slots – felt when you turn an unpowered motor by hand.
- Repeats every $360^\circ/\mathrm{LCM}(\mathrm{slots}, \mathrm{poles})$.

Example

12 slots, 8 poles: LCM $= 24 \Rightarrow$ period $360^\circ/24 = 15^\circ$ mech.

- Cure: measure i_q vs. angle at slow speed, store in a **look-up table**, add it to i_q^*.
- Matters most at low speed.

Check your understanding – 3.6

Check your understanding

1. A vertical axis must hold a load against gravity. Which compensation would you add?
2. Explain in one sentence how a disturbance observer "sees" the load torque.
3. What is the cogging period of a 9-slot / 6-pole motor?

Discuss with your neighbour for 2 minutes.

Summary 3.6 – Disturbances and compensation

Key take-aways

- Load torque and friction push against the motor; PI fixes them slowly.
- Feedforward cancels *known* effects from the plan.
- A DOB estimates the *unknown* push from $k_t i_q - \hat{J}\dot{\omega}$.
- Cogging repeats with angle $\Rightarrow$ table compensation.

Key terms

Load torque T_L – external push on the shaft.
Stick-slip – jerky motion from friction.
DOB – disturbance observer.
Cogging – magnet–slot torque at zero current.
Look-up table – stored correction vs. angle.

Session 6

Day 4 Wrap-up

Day summary

From voltage to motion

3.2 PWM sets an average voltage; SVPWM reaches $V_{\text{dc}}/\sqrt{3}$; dead time, f_{sw} and the limits shape what is possible.

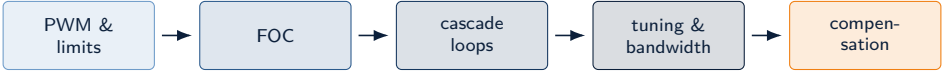
3.3 FOC: Clarke + Park (rotating camera) turn AC into DC; torque $= k_t i_q$.

From loops to accuracy

3.4 Cascade P–PI–PI with simple recipes, anti-windup, S-curves.

3.5 Bandwidth, inside-out tuning, 3–10× separation, delay cost.

3.6 Feedforward and observers remove what feedback alone cannot.



Preview of Day 5

Resonance, Sensors, Measurement and System Integration

- 3.7 Resonance, vibration and filters – continuing today's swing-like resonance and notch.
- 4.1–4.2 Sensors; encoders and resolvers – where θ_m and ω come from.
- 4.3–4.4 Sampling and signal processing – quantisation, aliasing, filter delay.
- 4.5 Communication and system integration.

Bring along from today
– the delay cost $360^\circ f_c T_d$,
– the velocity-loop plant $k_t/(Js)$,
– the two-mass FRF.

Figure credits

All plots, block diagrams, vector diagrams and schematics without a credit line are original drawings computed for this course (data generated by `day4/src/gen_data.py`). Downloaded figures (Wikimedia Commons):

File	Source (Wikimedia Commons) and author	Licence
d4_igbt_module.jpg	File:CM600DU-24NFH.jpg – Pslawinski	CC BY-SA 3.0
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